MULTI-STAGE AI PROMPT WORKFLOW

Hostel Complaint Priority Prediction and Escalation

Assessment Submission

1. Scenario Selected

The selected scenario is a Hostel Complaint Priority Prediction and Escalation workflow. The objective is to transform unstructured hostel complaints into a structured priority plan, draft an operational response, and then use AI to evaluate and refine the response.

The workflow also considers repeated complaints about the same issue and delayed resolution. Repeated complaints increase the urgency because they indicate that the issue may be affecting multiple students. If a complaint remains unresolved beyond the defined time limit, the manager should receive an escalation notification.

2. Objective

- Classify hostel complaints into suitable priority levels.
- Identify repeated complaints and use them as an urgency signal.
- Generate a practical action plan for hostel management.
- Escalate complaints that remain unresolved beyond the allowed time.
- Use a three-step chained AI workflow with a mandatory self-critique stage.

3. Input Data Used

Complaint ID	Complaint	Location	Repeated?	Time Open
C001	No water supply in the hostel bathroom.	Block A	Yes – 8	2 hours
C002	Tube light not working in room.	Block B	No	1 hour
C003	Food quality is poor and several students reported it.	Mess	Yes – 15	5 hours
C004	Water leakage near electrical switchboard.	Block C	Yes – 4	3 hours
C005	Wi-Fi is slow in the study room.	Block A	Yes – 10	6 hours

4. Prompt Chain Transcript and Intermediate AI Responses

Step 1 – Extraction and Planning

Prompt 1 – Exact prompt used:

> *Act as an expert hostel management analyst. I will provide a set of hostel complaints. Analyze them and extract the key factors that should determine complaint priority. Consider safety risk, number of*

affected students, repeated complaints, type of problem, and how long the complaint has remained unresolved. Create a structured priority framework and apply it to the complaints. Also identify which complaints should trigger manager escalation if they remain unresolved. Present the result as a table with Complaint ID, Priority, Reason, Suggested Action, and Escalation Condition.

Input complaints: C001 No water supply in Block A bathroom; C002 tube light not working in Block B; C003 poor food quality reported by several students in the mess; C004 water leakage near an electrical switchboard in Block C; C005 slow Wi-Fi in the study room.

Intermediate AI Response – Step 1:

ID	Priority	Reason	Suggested Action	Escalation Condition
C004	Critical	Electrical safety risk combined with water leakage and repeated reports.	Restrict the area and send maintenance/electrical staff immediately.	Escalate to manager if not attended immediately.
C001	High	Essential water service is unavailable and the issue is repeatedly reported.	Send plumbing/maintenance staff and restore water supply.	Escalate if unresolved within 4 hours.
C003	High	Food quality affects many students and has repeated reports.	Inspect food preparation and mess operations.	Escalate if unresolved within 6 hours.
C005	Medium	Repeated issue affects study activities but is not an immediate safety threat.	Check network equipment and bandwidth.	Escalate if unresolved within 12 hours.
C002	Low	Single lighting issue with limited immediate impact.	Replace or repair the light.	Escalate if unresolved within 24 hours.

Step 2 – Generation / Drafting

Prompt 2 – Exact prompt used:

Using only the priority framework and complaint analysis you produced in Step 1, create a hostel management response plan. Do not introduce new complaint categories. The plan must explain how complaints are handled in priority order, how repeated complaints increase urgency, and how unresolved complaints are escalated. Include a simple notification message that can be sent to a hostel manager when an escalation condition is reached. Keep the response practical and suitable for a student hostel management system.

Intermediate AI Response – Step 2:

The system should process complaints in the following order: Critical, High, Medium, and Low. Safety-related complaints receive the highest priority. Complaints reported repeatedly by several students should receive increased urgency because they indicate a wider impact. Essential services such as water and food should be handled before lower-impact convenience issues.

For each complaint, the system records the complaint time, current status, priority, number of similar reports, assigned staff member, and resolution time. The system checks unresolved complaints against their escalation condition. When the allowed time is exceeded, an automatic notification is sent to the hostel manager.

Example escalation notification:

> **ALERT: Complaint C001 has exceeded its 4-hour resolution limit. Issue: No water supply in Block A bathroom. Repeated reports: 8. Priority: HIGH. Please review the complaint and assign immediate action.**

Step 3 – Evaluation and Refinement (Critique Gate)

Prompt 3 – Exact prompt used:

> *Before finalizing the response plan, act as a strict senior reviewer. Evaluate the Step 2 draft against this checklist: (1) Does it clearly prioritize safety and essential services? (2) Does it explicitly use repeated complaints as an urgency factor? (3) Does it explain what happens when a complaint is not solved within the specified time? (4) Does the manager notification contain enough information to act? (5) Is the workflow practical for a hostel complaint management system? First list any weaknesses you find. Then produce a corrected final version that fixes those weaknesses without changing the core information from Step 1.*

Intermediate AI Response – Step 3 Critique:

- The draft correctly places safety and essential services ahead of lower-impact complaints.
- Repeated complaints are mentioned as an urgency factor, but the final workflow should make this rule more explicit.
- The escalation process is present, but the final version should clearly state that the system checks unresolved complaints against their time limits.
- The notification contains the essential complaint information, but it can be made more actionable by including priority, repeated-report count, and elapsed time.

Refined Final Output:

The proposed hostel complaint management workflow first classifies complaints according to safety risk, service importance, number of affected students, repeated reports, and time remaining before escalation. A complaint involving an immediate safety risk is treated as Critical. Essential services such as water and food are treated as High when they affect students or are repeatedly reported. Medium and Low priorities are used for issues with less immediate impact.

Repeated complaints are treated as an additional urgency signal. For example, if many students report the same water, food, or Wi-Fi problem, the system identifies the repeated issue and gives it greater attention than an isolated complaint of the same general type.

After assignment to maintenance or another responsible staff member, the system continuously checks whether the complaint is still unresolved. If the configured resolution time is exceeded, the system automatically escalates the complaint to the hostel manager.

Final Manager Notification Example: ALERT – Unresolved Hostel Complaint. Complaint ID: C001. Issue: No water supply in Block A bathroom. Priority: HIGH. Similar reports: 8. Time open: 4+ hours. Status: Unresolved. Action required: Please review and assign immediate resolution.

5. Final Workflow Summary

Stage	Input	AI/ System Action	Output
1. Extraction / Planning	Raw hostel complaints	Identify priority factors and classify complaints.	Priority framework and complaint table
2. Generation / Drafting	Step 1 output	Create management response and escalation process.	Operational response plan + notification
3. Evaluation / Refinement	Step 2 output	Review against checklist and correct weaknesses.	Final polished workflow

6. Expected System Output

- A ranked list of hostel complaints based on urgency.
- A priority label such as Critical, High, Medium, or Low.
- Detection of repeated complaints about the same issue.
- Assignment of complaints to the responsible maintenance or hostel staff.
- Tracking of unresolved complaints and elapsed resolution time.
- Automatic manager notification when a complaint exceeds its escalation limit.
- A dashboard that allows management to see urgent and overdue complaints first.

7. Evidence of Live Conversation

The assessment rules require direct evidence of the live conversation thread in the AI interface. Insert the screenshot(s) of the actual three-prompt chat below this heading before submission.

[INSERT SCREENSHOT OF THE LIVE 3-STEP AI CHAT HERE]

8. Compliance with Assessment Rules

Requirement	Status
At least 3 chained prompts	Satisfied – three sequential prompts are documented.
Each prompt depends on previous output	Satisfied – Step 2 uses Step 1; Step 3 reviews and refines Step 2.
Critique gate	Satisfied – Step 3 explicitly evaluates the previous draft against a checklist.
Document exact prompts	Satisfied – all three prompts are included verbatim in this submission.
Intermediate responses	Satisfied – outputs from each stage are documented.
Live chat evidence	Screenshot must be inserted in Section 7 from the actual AI interface.

9. Conclusion

This multi-stage AI workflow demonstrates how a hostel complaint management problem can be handled through chained prompting rather than a single request. The first stage structures the problem, the second stage converts the analysis into an operational plan, and the third stage critically evaluates and improves the result. The proposed approach also addresses repeated complaints and delayed resolution by using them as signals for increased urgency and manager escalation.